Fiscal Year:	FY 2013	**Task Last Updated:**	FY 02/19/2013
PI Name:	Morris, Christopher James Ph.D.		
Project Title:	Cardiovascular and Metabolic Consequences of Circadian Misalignment		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 NSBRI-RFA-10-01 Postdoctoral Fellowships
Start Date:	11/01/2010	**End Date:**	10/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Change to end date from 11/30/2012 to 10/31/2012, per NSBRI (Ed., 9/22/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Scheer, Frank (MENTOR/Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-PF02303		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Original project aims/objectives: The risk of adverse cardiac events has been listed as a major priority in the NASA Bioastronautics Roadmap. 'Slam shifts', which entail rapidly changing an individual's sleep/wake cycle, are frequently performed by astronauts. Epidemiological research indicates that slam shifting is associated with adverse cardiometabolic function. The possible adverse cardio-metabolic consequences of slam shifts could significantly harm an astronaut's performance and therefore the likelihood of mission success. The association between slam shift work and health problems may be due to circadian misalignment. However, the limitation of any epidemiological study is that the underlying mechanisms cannot be deduced with any degree of certainty. Thus, this calls for an experimental study to investigate the effect of circadian misalignment per se and to assess in detail potential underlying mechanisms for the increased health risk with slam shift work. Our primary aim is to determine the effect of simulated slam shift work on physiological changes that increase susceptibility to the development of cardiovascular disease. This will be accomplished by comparing cardio-metabolic, vascular, endocrine and inflammatory changes when subjects are scheduled to a 3-day simulated slam shift schedule versus a 3-day simulated day shift schedule. To address the question of whether or not individuals adapt to repeated slam shift exposures, we aim to study a control group of day workers and a group of chronic slam shift workers in an identical experimental. Our independent variables include: (a) circadian alignment (misaligned [slam shift] vs. aligned [day shift]); (b) shift duration (first day vs. 3rd day on slam shift or day shift); (c) chronic slam shift workers vs. non-slam shift workers. Key findings: To date, we have successfully completed the study of 13 subjects that each undertook two 8-day in-laboratory stays separated by 2-12 weeks. We are still in the process of receiving results (e.g., cortisol, leptin, ghrelin, lipids, etc.) for the samples from all subjects who have completed the study. Below are preliminary findings from the study. Here, we only show results pertaining to mixed meals consumed on Day 1 of day work and night work. Fasting glucose levels were similar between conditions (mean±SEM: day work, 85±2 mg/dl vs. night work, 85±2 mg/dl; P=1.00). During night work, glucose 3-h area-under-the-curve (AUC) was 23% higher (342±17 mg/dl.3 h vs. 419±13 mg/dl.3 h; P<0.0001) and peak glucose levels were 16% higher (166±7 mg/dl vs. 193±4 mg/dl; P<0.0001). There was a significant interaction of condition*time-since-meal (P<0.0001), with slower decline of glucose levels during night work. Fasting insulin levels were 15% lower during night work (5.1±0.8 uIU/mL vs. 4.3±0.7 uIU/mL; P=0.035). Insulin 3-h AUC was not significantly different but peak insulin levels were 13% lower during night work (99±15 uIU/mL vs. 86±13 uIU/mL; P=0.035). For insulin, there was a significant interaction of condition*time-since-meal (P<0.0001); with 33-41% lower levels at min 20 (P=0.002) and min 30 (P=0.002), but 40-50% higher levels at min 80 (P=0.005) and min 90 (P=0.025), possibly due to the elevated glucose values during night work. More preliminary findings are presented in an Appendix submitted to NSBRI. Impact of key findings: It is vital that we understand the adverse health effects of spaceflight (e.g., circadian misalignment caused by slam shifts). This will ensure that countermeasures are developed to limit or alleviate the adverse health consequences of spaceflight and thus ensure mission success. Moreover, approximately 8.6 million Americans undertake some type of slam shift work and consequently the findings from this project will be applicable to ground-based people as well as astronauts. Research plan for the coming year: We aim to complete data analysis for the control group and finish studying the shift worker group.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Approximately 8.6 million Americans undertake shift work routines similar to those experienced by astronauts on the ISS. Thus, the findings regarding adverse affects of shift work on cardiovascular and metabolic factors from the current project are applicable to both astronauts and people on Earth. Our findings will enable better countermeasures to be developed which will prevent or attenuate the adverse health effects of shift work.
Task Progress:	Over the last year, we have completed 11 new studies, totaling 176 bed days, or about 4,324 hours of intensive physiological monitoring under controlled laboratory conditions. Recruitment has picked up and gained momentum now that we have a more practical protocol, less restrictive inclusion criteria, and focused recruitment strategies to target the specific subject population, as detailed in our previous Progress Report. We are now in the process of analyzing collected data in order to provide preliminary data for future grant applications. Over the coming months, we plan to analyze blood samples for cardiovascular and metabolic factors. Measures include tPA activity, PAI-1, post-prandial glucose, insulin, and c-peptide concentrations, lipids, inflammatory makers, leptin, ghrelin, and resting metabolic rate. In summary, by the end of next year, we will have a substantial data set that will provide novel and clinically relevant insights into the effect of slam shifts on cardiovascular and metabolic health. Furthermore, since the last progress report I have published two reviews (Progress in Brain Research, Molecular and Cellular Endocrinology), two original papers (Obesity and Sleep), and have another manuscript under review (American Journal of Hypertension). All these manuscripts are partly attributed to this grant.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
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